

# Generation Interconnection Combined Feasibility / System Impact Study Report

## For Queue Project Q18-0929

Disputanta 138 kV  
June 26, 2019

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0929. The Interconnection Customer proposes a 303.8 MW wind facility interconnecting at the Disputanta 138 kV substation in the Virginia area. The request was received on January 23, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 345 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$24,621,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                        | Contingency                          | Loading (%) | Rating (MVA) | DFAX (%) |
|-------------------------------------------|--------------------------------------|-------------|--------------|----------|
| Disputanta – Wagley Robertson 138 KV line | Disputanta – Floraday 138 KV         | 167.2       | 170          | 100.0    |
| Floraday – Disputanta 138 KV              | Wagley Robertson – Disputanta 138 KV | 141.2       | 202          | 100.0    |

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility                                  | Required Upgrade                            | Cost Estimate | Allocation to Project |
|-------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Disputanta – Wagley Robertson 138 KV line | Rebuild line with higher-capacity conductor | \$13,038,000  | 100% (\$13,038,000)   |
| Floraday Substation – Disputanta 138 KV   | Rebuild line with higher-capacity conductor | \$11,583,000  | 100% (\$11,583,000)   |
|                                           | Total allocated to project                  |               | \$24,621,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.